

SPE-229755-MS

From Conventional to Electric: Advancing Tanker Efficiency and Safety in the IMO 2050 Era

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229755-MS](https://doi.org/10.2118/229755-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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In this article we are focusing on the cargo pump systems for tanker vessels. The study applies to IMO Type I, II and III tanker vessels in the size of Handy, MR and LR1 class (5.000-85.000 DWT) with a deep well tank design, which implies an in-tank cargo pump for each cargo tank. The pump design is a centrifugal pump.

The usage of this pump system is for liquid cargoes as specified in the **International Bulk Chemical (IBC) Code**, developed by IMO. This code classifies chemical cargoes into three ship types—**Type 1, Type 2, and Type 3**—Type 1 posing the most severe hazards. The cargoes associated with each type are listed in **Chapter 17** of the IBC Code.

The cargo pump system is critical equipment of a tanker vessel.

Introduction to this topic

In this article we are focusing on the cargo pump systems for tanker vessels. The study applies to IMO Type I, II and III tanker vessels in the size of Handy, MR and LR1 class (5.000-85.000 DWT) with a deep well tank design, which implies an in-tank cargo pump for each cargo tank. The pump design is a centrifugal pump.

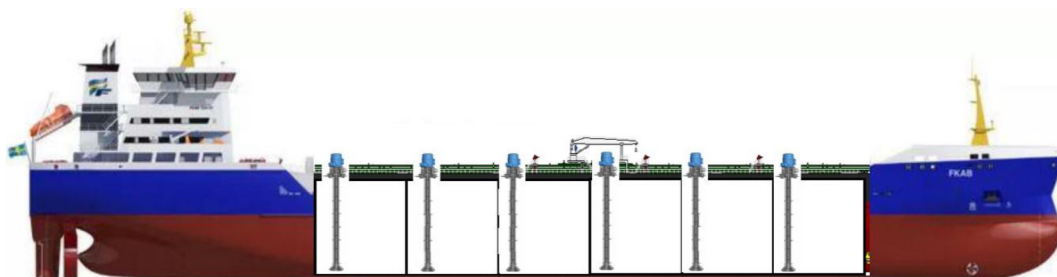


Figure 1—lay out of deepwell design tanker vessel

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The cargo pump system is critical equipment of a tanker vessel.

1. **Primary Role in Cargo Operations:**

- They are essential for **discharging cargo**. Without functioning pumps, cargo operations cannot proceed efficiently or safely.

2. **Safety and Environmental Protection:**

- Malfunctioning pumps can lead to **spills, overflows**, or **residual cargo** remaining in tanks, posing **environmental hazards** and **safety risks**.

3. **Compliance with Regulations:**

- SOLAS, MARPOL, and the IBC Code require that cargo handling systems, including pumps, be **maintained in good working order**.

A categorized overview of the technological eras from 1950 to today, focusing on cargo pump systems and broader maritime engineering trends

Era	Time frame	Technical focus (Cargo pumps)	Business Focus
Mechanical Era	1950 - 1970	Central pump rooms Manual Operations Gear and Screw pumps	Focus on tonnage and fleet expansion Low Cost Operations
Hydraulic Innovation Era	1970- 1990	Introduction of hydraulic deepwell pumps/One pump per tank concept Electric motors for auxiliary equipment	Safety and segregation improvement
Digital Transition Era	1990- 2010	Rise of electric driven deepwell pumps Variable Speed Drive system introduced Split Matrix system	Compliance with IMO Focus on reliability
Smart Integration Era	2010- 2025	Electric pumps with full digital control High efficiency e-motors Hydraulic systems challenged to comply with sounds emission and pollution Full Drive systems and Smart Pumping	Energy efficiency ESG and sustainability goals
Autonomous and Business driven Era	Future	Electrical systems become the standard Intelligent control systems Autonomous operations?	Data - driven decision-making Performance based contracting Comply with IMO2050

Hydraulic Driven Cargo Pump Systems: Mechanical Era Excellence

Hydraulic deepwell pumps, were introduced in the late 1960s, ideally suited to the mechanical era, which emphasized **robustness, direct power transmission, and mechanical reliability**. They offered significant operational and vessel design improvement compared to pump room systems.

These systems became deeply embedded in tanker design and operations, making them the default choice for many shipowners and operators.

The Rising Demand for Electric-Driven Deepwell Cargo Pumps

Changing **market requirements** — driven by environmental regulations, profitability, energy efficiency goals, and digitalization — are significantly increasing interest in **electric-driven cargo pumps**. Here's why:

- Same benefits as the conventional standard Deepwell cargo pumps.
- Coherent with the digital age transformation.

Transition in Progress

Tanker newbuilds and retrofits are increasingly considering electric alternatives. The shift is not just technological — it's strategic, aligning with broader goals of **sustainability, compliance, and operational efficiency**.

In new regulations such as EU ETS and CII, the cargo pump system does affect results in efficiency of vessel performance and in specific regulations, such as the CARB regulation in California, the main contributor for a higher energy efficiency rating is the cargo pump system. And nowadays this has been acknowledged by Class societies such as DNV. DNV has published an article about vessel improvements with recommendations.

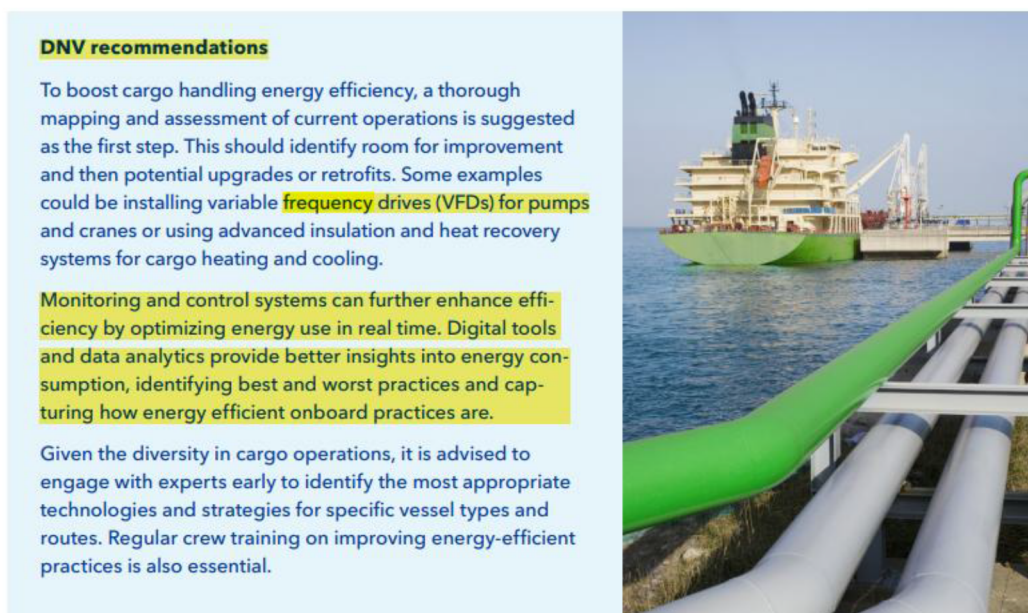


Figure 2—DNV Energy efficiency and measures technologies, <https://www.dnv.com/maritime/publications/energy-efficiency-report-download/>

In this article, advancing tanker efficiency in the IMO 2050 era, will be approached from a technical and operational (safety) perspective specifically for the cargo pump systems.

To create more awareness of systems that are available and how this impacts the efficiency rating of the tanker fleet for shipowners, decrease CO₂ emissions, increase the safety and comply with future regulations towards IMO 2050.

Advancing tanker efficiency from a technical perspective

As mentioned in the introduction, the hydraulic and electric driven pumps systems are similar in design, both are centrifugal pumps with similar operational features for the discharge operations and to comply with Class requirements. The main difference is the system being used for driving the (centrifugal) cargo pumps.

For the hydraulic driven system, it has the following flow: from the generator set, the hydraulic unit is powered and generates the hydraulic pressure that goes via hydraulic high-pressure piping to each cargo pump. Each cargo pump has a hydraulic motor to power the centrifugal cargo pump.

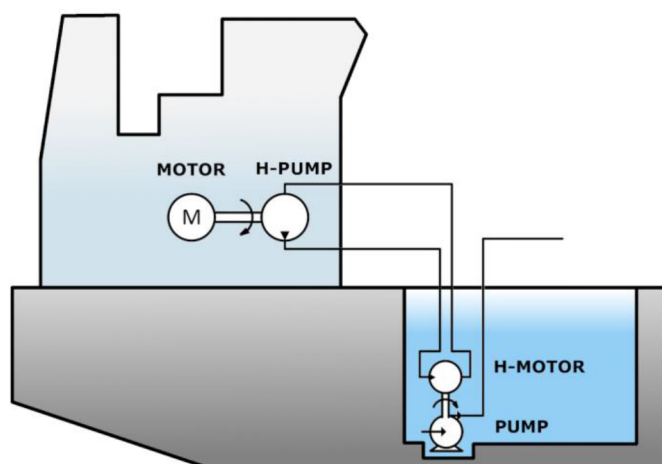


Figure 3—schedule of Hydraulic driven pump system

Gen set → hydraulic unit → high pressure hydraulic piping → hydraulic motor → cargo pump

The electrical pump system has a similar flow but instead of high-pressure hydraulics, electrical power is used for driving the centrifugal cargo pump. The main switchboard sends electric power to the frequency converter system. With electrical cables the power goes to the e-motor that is on top of the deck and the e-motor powers the centrifugal cargo pump.

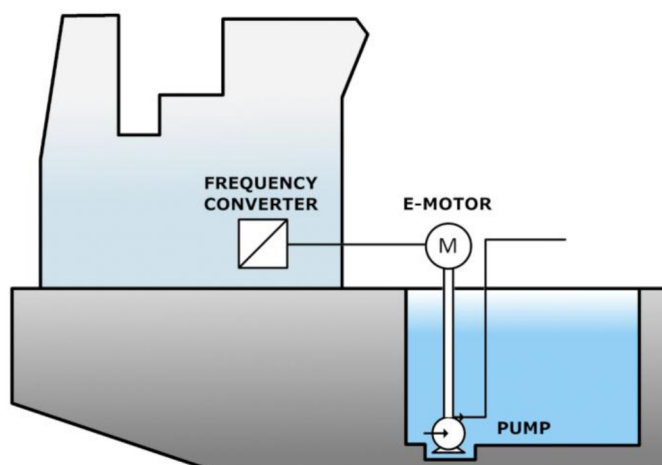


Figure 4—schedule of Electric driven pump system

Main switch board → Frequency converter → electrical cables → e-motor → cargo pump

In design and operational perspective, both hydraulic and electric driven cargo pump systems are very similar. The same counts for the discharge capacities ranges and pump curves.

In many maritime companies, where the crew rotates over the different vessels in their fleet where they have both hydraulic and electrical driven systems, the feedback on operational handling between both is considered similar.

However, there are major differences in certain aspects.

13-18% higher energy efficiency for an electric driven cargo pump system (without smart pump technology).

In the hydraulic system, there are losses of (hydraulic) energy from the hydraulic unit, via piping to the hydraulic motor/pump.

Where an electrical signal goes through a secured power cable has minimal power loss.

This is a given by default of technology and is not related to any manufacturer of a pump system.

Noise level. The hydraulic pump system has a high pitch noise level caused by the hydraulic unit, the flow controller, piping in combination with high pressures. Crew and the environment, including ports in rural areas, Rotterdam & Antwerp to name a few, can become hindered by this noise. Dampeners and additional noise limiting equipment can be applied to reduce noise levels of hydraulic systems. This is not needed for electric driven systems. The electric driven pump system is quiet in operation with a noise level of less than 60 dB. (1 meter away from the electric motor that drives a pump)

Companies that have a fleet that consist of both electric as hydraulic driven systems and where the crew rotates on the vessels with both systems, by the crew it is acknowledged and confirmed that the electric driven system is more pleasant working in terms of noise pollution.

In Europe there are some ports where discharging with hydraulic driven cargo pump systems is restricted at certain hours.

The trend towards the future is that noise emissions will be considered to become more important, for the wellbeing of the crew and environment, for both nature as human beings, as residential areas are expanding towards terminal areas.

Another major difference is the size of the total equipment. Outside the cargo area, an electric driven system requires less machinery, the electric driven system saves between 20-60 M2 of space compared to a hydraulic driven system.

Lastly, and this is also important for the future of shipping, the hydraulic driven system is a mechanical system, by design not smart and it lacks the data and capability to become smart. Where the electric driven system is driven by software and generates an abundance of data useful for data driven decision making and intelligent automation. And by interfacing the system to other digital systems on board, it can start to make data connections and digital twin analytics to add extra equipment to receiving feedback and measuring data, such as electrical current of the pump while operating, it gives possibilities to connect with other existing equipment related to the discharge operations. All these signals are electric signals that could be integrated into the software to operate the electric driven cargo pumps and improve efficiency in its operation.

In the trend of electrifying equipment to save energy and to decrease emissions, it gives the opportunity to make the discharge process more dynamic. With many operational benefits for the crew in terms of energy efficiency, safety and ready for IMO 2050.

Digital Twin Analytics: Electric vs. Hydraulic Cargo Pump Systems

Electric-driven cargo pump systems offer distinct advantages over hydraulic systems when it comes to digital twin analytics. These benefits stem from superior data accessibility, standardized communication protocols, and simplified modeling.

1. **Sensor Integration and Data Availability**

Electric systems typically include built-in sensors for motor current, voltage, RPM, temperature, and vibration. This structured, high-frequency data is ideal for real-time monitoring and predictive analytics. Hydraulic systems, by contrast, rely on pressure, flow, and oil condition sensors, which are less standardized and often require custom instrumentation.

2. **Modeling Simplicity**

Electric motors follow well-defined physical models (e.g., torque-speed curves), making them easier to simulate and predict. Hydraulic systems involve complex fluid dynamics, which are nonlinear and temperature-dependent, complicating accurate modeling.

3. **Connectivity and Control**

Electric pumps are commonly controlled via Variable Frequency Drives (VFDs) and PLCs, which are digitally native and easily integrated into digital twin platforms. Hydraulic systems often use analog control loops or centralized HPUs, requiring additional digital interfaces for integration.

4. **Modbus Interface Advantages**

Electric systems frequently support Modbus RTU or TCP/IP protocols, enabling standardized communication with SCADA systems and digital twin platforms. This facilitates real-time data access, remote diagnostics, and scalable integration. Hydraulic systems typically lack native Modbus support, necessitating additional hardware and configuration for similar functionality.

5. **Predictive Maintenance and Scalability**

Electric systems support condition-based monitoring using motor signature analysis and fault code tracking. Their modular architecture and digital interfaces make them easier to scale and maintain. Hydraulic systems require fluid analysis and mechanical wear tracking, which are more complex and less compatible with automated analytics.

Study on discharging

The goal for discharging is to ensure a safe operation, discharge according to terminal and charter parties and compliance with regulations.

The financial benefit of efficiency improvements in tanker operations depend on the charter party structure.

Charter Type	Default ETS Cost Payer	Notes
Time Charter	Charterer (by agreement)	Charterers often pay for fuel and voyage costs, so ETS costs are typically passed to them via clauses like BIMCO's ETSA Clause.
Voyage Charter	Owner (default)	Owners are responsible for emissions and surrendering allowances, but may include ETS costs in the freight rate.
Bareboat Charter	Charterer	The bareboat charterer is considered the "shipping company" and bears full operational and compliance responsibility.
COA	Depends on contract	ETS cost allocation is negotiated and may vary per voyage or across the contract term.

As a result, efficiency improvement is not always a key driver for crew on board

For energy and time saving of cargo discharge operations and port call optimization, many variables come into play. From a pump perspective it is the ideal combination of pump speed, pressure and flow. In addition, the viscosity and specific gravity of the cargo needs to be determined. But there are external influences, not directly related to the pump equipment, that influence a most efficient discharge, such as tank levels that change, inlet pressure, discharge valve positions, and terminal layout and infrastructure.

In the following graph, an example of a certain discharge has been visualized.

On the left Y-axis, is the pressure in Head meters

On the X-axis, the flow in M3/h

On the right Y-axis, is the pump speed

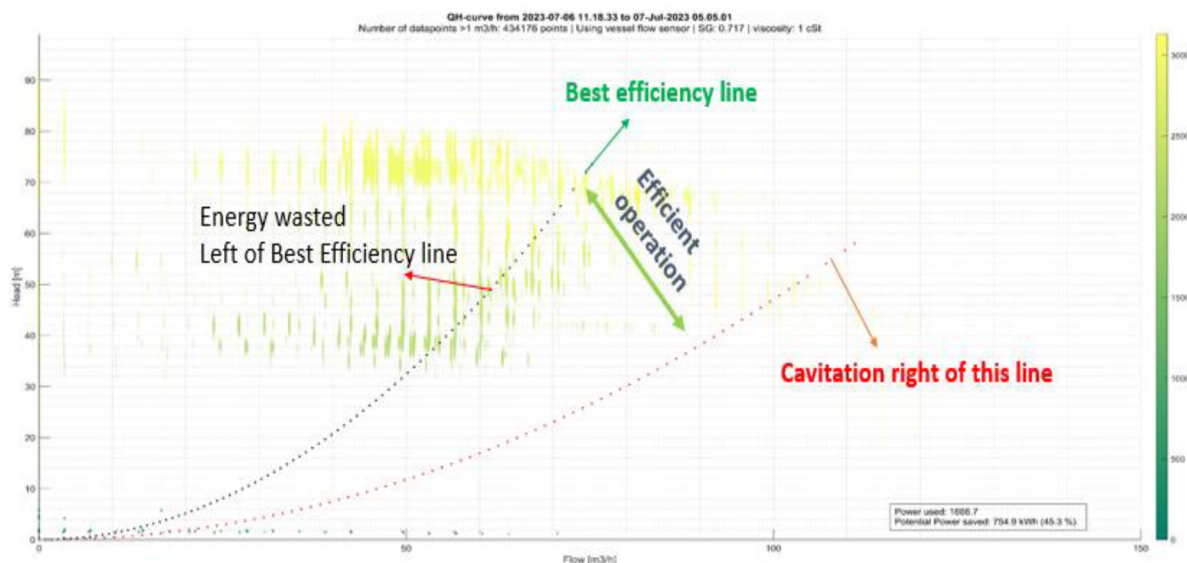


Figure 5—efficiency chart of a centrifugal pump

Regardless if the centrifugal pump is hydraulic or electric driven, centrifugal pumps all have the same generic characteristics.

They are good at discharging a wide range of cargoes from light to heavy and fluid or less fluid liquids. They all have similar pump curves and behaviours. With cargo Specific Gravity (SG) and Viscosity, and the ideal combination of pressure (head), flow and pump speed, an Operational Excellence Area is available. Shown as the area "Efficient operation" between the green and red line in the graph. That means operating the pump for discharging anywhere in that area would be the most desirable for that certain cargo.

If the pump operates on the left side of this line, it means unnecessary energy is being used for the discharge. So, more kWh's are used for the discharge that could have been within the Operational Excellence Area. This can happen for example when a discharge valve is not in the optimal position, or the pump speed is not adequate to reach that duty point.

There is a red line, which indicates cavitation danger. In a situation where the cargo pumps operate on the right of this line, it means there is cavitation. Later in this article we will go further in detail on cavitation.

During operations of the pumps, the duty point is where the pump is operating, this can be at any point in the graph. Seen in the graph as the yellow/green could in the back, these are actually duty points. Each second the pump is in operation; a duty point will be registered.

In the most ideal situation, the entire discharge process will happen within the Operational Excellence area. But manually this is impossible to reach. As mentioned, the crew is facing external factors that influence this Operational Excellence Area, such as tank levels that change throughout the discharge process

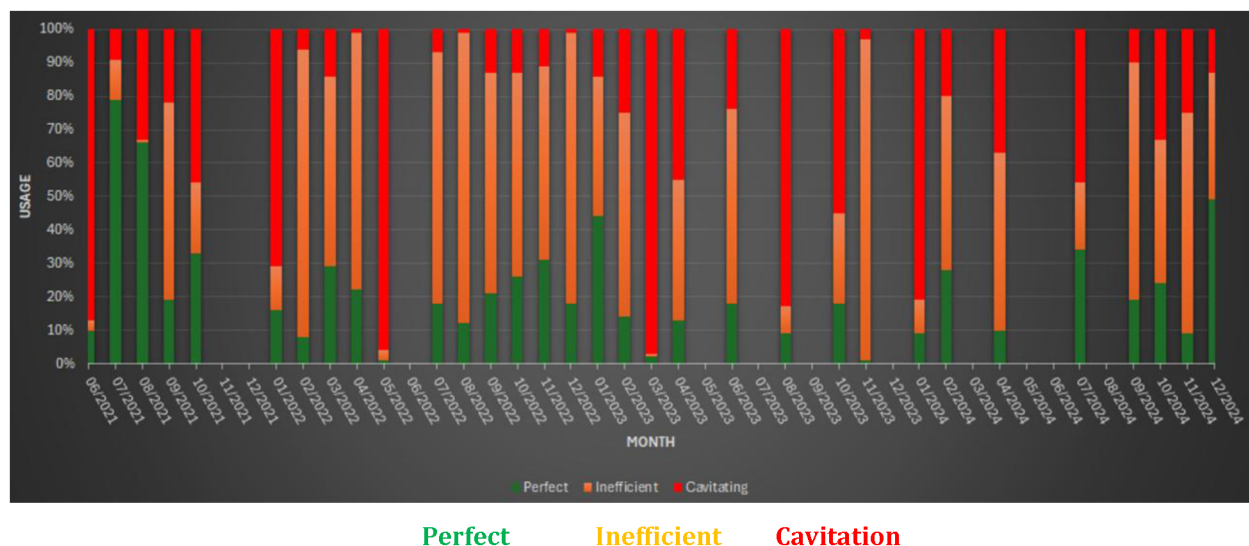
and changing tank levels has an influence on the pressure. Different discharge valve positions for each cargo tank and others.

Most important, for the crew it is humanly impossible to observe all these variables during the discharge process as it happens below deck, there are no visuals in piping, pumps etc and as it is dynamic, it changes throughout the discharge process and in each tank differently.

In a study with LR1 tankers from worldwide largest renown tanker owner, a digital twin has been installed to measure performance during discharging. During the study, the discharge processes have been measured and analyzed over >3-year period. Monitoring and storing various data.

By logging the pump operations with the cargo specs, the discharge could be visualized in the efficiency graph and analyzed in which duty points the cargo pumps have operated.

The results were clear that there is a huge space for improvement to discharge the cargo more energy efficiently and without cavitation. Hereunder the results where Green is the perfect discharge area, the Operational Excellence Area, Orange is energy inefficient usage (left of Best Efficient Line in the graph) and Red is risk on cavitation, right of the cavitation line.



In the earlier explanation of the difference between hydraulic and electric cargo pump systems, it was mentioned that the benefit of an electric driven pump system is that the pumps can be driven by a Smart Pumping software in the frequency drive system. This is not applicable in hydraulic driven pump system as it is mechanically driven.

The electric driven system gives the possibility to connect with other existing equipment on board to receive input, such as tank levels, discharge – manifold pressures and discharge valve position. With this input and the determination of the cargo specs, the cargo pump can operate in the Operational Excellence Area automatically. And during the discharge process, adjust speed and/or valve position when needed to ensure the discharge process remains in the Operational Excellence Area throughout the entire discharge process.

In comparison with using the Smart Pumping software, it has been proven that the energy savings can be up to 45% for a similar discharge process without the Smart Pumping software, where the discharge process is done manually. This is on top of the energy saving of using an electric driven pump systems versus hydraulic as mentioned in this article earlier.

The Smart Pumping software is an intelligent software that enables automatic discharge within the Operational Excellence Area.

The crew sets in the cargo specifications, the Specific Gravity and Viscosity. Enter the discharge rate and pressure at manifold, according to the charter contract or regulation and then select the tanks to discharge.

The Smart Pumping starts initiate the pumps and through the Mod Bus communication with other existing equipment, the discharge will be controlled to stay within the Operational Excellence Area. The software has the feature to log all data during the discharge and calculate exact power consumption.

The power saved calculation:

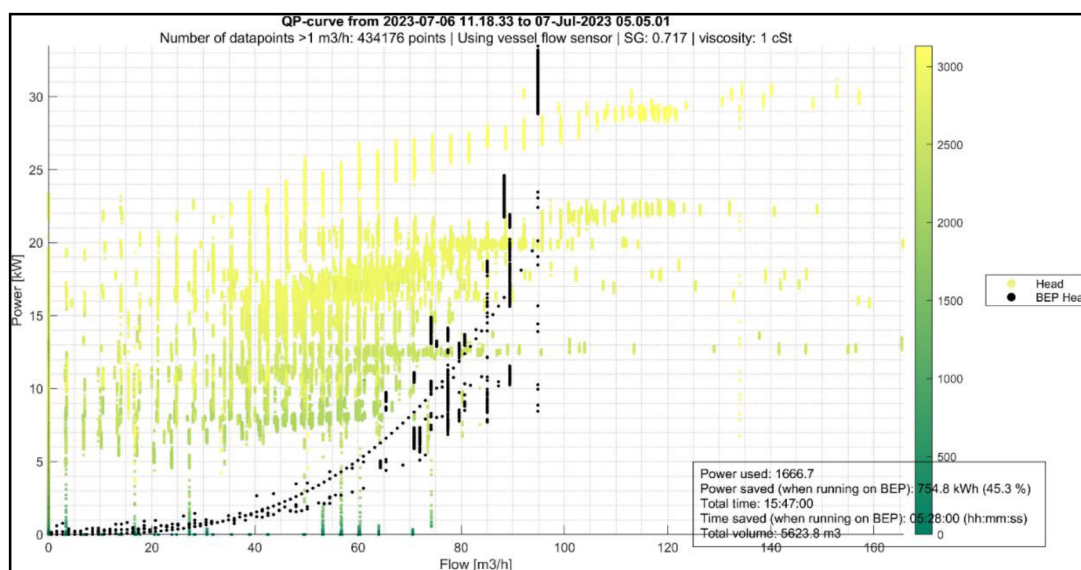
For every point left of the Operational Excellence Area, we have made a calculation to derive the power that would be needed to have the same head but with the flow corresponding to the Operational Excellence Area. This point was probably chosen by the pump operator because a certain head was needed therefore we keep the same head make sure the flow will always be more than the original duty point indicated.

So for instance we take a point of 80 m head at 40 m³/h, this results in a pump speed of 3115 rpm. The required pump power will be 16.6 kW.

To get the same head but in the Operational Excellence Area the new point becomes 80 m @ 86 m³/h with a speed of 3345 rpm. The required pump power will be 27.9 kW on this point.

This means that the pump would be using 68% more power but would also generate 115% more flow for the same head.

For the power the same sort of graph can be made:



Say the tank is 800 m³, if the pump had run on this point constantly it would have taken $800/40=20$ hours to empty the tank and it would have cost $16.6*20=332$ kW to do so.

If the pump had run in the Operational Excellence Area it would have taken $800/86=9.3$ hours and it would have cost $27.9*9.3=259.5$ kW. The total savings if only these two points would have been used for the whole unload would have been 28% energy and 10.7 hours of unloading time.

For the "Power saved" calculation every datapoint (every second) located left of the Operational Excellence Area curve was recalculated as if it were moved to the Operational Excellence Area keeping the same head but changing the flow.

At last, in terms of future developments, we will see Shore power to become more available in ports across the world. And in certain areas, such as Californian ports under the CARB regulations, shore power will be even obligated if shore power connection is available on the vessel.

The electric power received from the terminal is not unlimited and has to be paid by the ship owner or ship manager. Therefore it is important to use the received shore power in the most efficient way to minimize energy consumption from the shore power.



Figure 6—Shore power



Figure 7—Full Drive system

While at the terminal, the cargo pumps are the major energy consumers of the vessel because the vessel will be discharging.

It is critical that the cargo pump system uses the shore power most efficiently possible.

From a technological standpoint, it is most efficient to use the received electric energy from shore directly to power electrical driven equipment. This way it uses the shore power most efficiently and to power the pumps directly. It is clear that an electric driven pump system is the only adequate solution to operate with shore power.

In case of a conventional hydraulic system. The received electrical energy from the shore must be converted to hydraulic power and this causes losses in efficiency, more kWh will be consumed for the same discharge.

In combination with the Smart Pumping software on the electric driven pump systems, the cargo pumps will operate within the Operational Excellence Area, avoiding unnecessary burning kWh for the same discharge.

- Electrical power in (from shore), electrical power out (e-motors) minimal energy loss
- Dynamically adaptation of the cargo pump speed and valve positions to maintain the pumps operating in the Operational Excellence area.

- Automatic reports of kWh consumptions

In the future, where the shore power can be generated fully from renewable sources. The complete discharge operation from supplying shore power to the electric driven equipment on board, fulfills a Net Zero impact and is aligned with the goals for IMO2050.

Safety from a technical perspective

Any centrifugal pump, no matter the drive system, has a risk on cavitation. With cavitation, shock waves/disturbance may damage the pump equipment and the construction of the tanks. During discharge, when the tank level lowers the Net Positive Suction Head available will also lower. When this NPSHa is lower than the NPSHr (required) the pump cavitates.

In larger tanker vessels, such as MR and LR1, where the pumps are larger with bigger impellers and larger capacities, it means larger and stronger shock waves/disturbance that leads to a higher impact of the cavitation in case cavitation occurs.

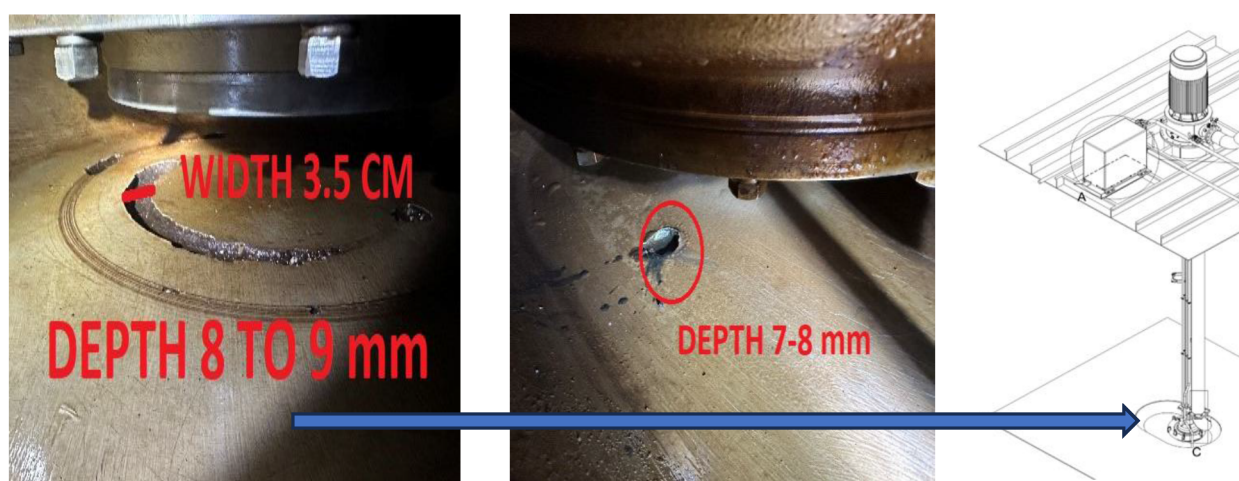


Figure 8—cavitation damage and location in the tank

The benefit from an electric driven system is that the Smart Pumping software is able to monitor the pressure. The system keeps checking the NPSHa and will make sure this will never fall below the NPSHr to prevent cavitation. In case there is a risk where the NPSHa will decrease, the pumps will automatically decrease speed and the valve will accordingly and gradually close, with the result that risk on cavitation has been avoided.

This feature of Smart Pumping software contributes to a safer operation for the crew and the lifespan of pump equipment and tank construction of the vessel.

Advancing tanker efficiency from a safety perspective in operations

In comparison, from 30 years ago to today, the crew is facing an increasing workload on procedures to avoid any incidents and to comply with regulations to operate as safely as possible.

Besides, more equipment on board has become ‘interactive’, not only cargo pump equipment but all other equipment on board as well.

The main goal for all equipment manufacturers is to operate their equipment in a safe and efficient way for the crew.

Due to the rapid technological developments of equipment, also more and more data is available for the crew. It is a challenge to filter the necessary data for their operations and that really contributes to an efficient

and safe operation. Manufacturers must be aware not to create an overflow of irrelevant information and neither the use of additional sensors to avoid having a vessel full with sensors with risks on sensor damage.

Smart pumping is only using communication between existing equipment; therefore no extra sensors are needed. There is always an option to overrule the software and operate the pumps manually when desired.

As described in the SIRE 2.0 regulations, we see that technology, software in relation to crew safety and wellbeing, goes hand-in-hand:

1. **Digitalization:**

- Inspections are now conducted using tablets with real-time data entry and photo documentation
- Vessel-specific data is uploaded in advance, improving inspection accuracy and transparency.

2. **Human Factors Focus:**

- Crew competence is assessed with ratings like *Not as expected*, *As expected*, etc.
- Performance Influencing Factors (PIFs) such as fatigue, morale, and ergonomics are considered to understand crew behavior and safety culture

3. **Improved Inspector Guidance:**

- Questions now include detailed guidance sections: objectives, expected evidence, suggested actions, and potential grounds for negative observations

Safety Impact

- **Holistic Safety Evaluation:** Combines hardware, procedures, and human competence.
- **Predictive Insights:** Enhanced data quality helps forecast future performance.
- **No Blame Culture:** Focuses on systemic issues rather than individual fault.
- **Continuous Improvement:** Reports guide corrective actions and training needs

Digitalization in **SIRE 2.0 inspections** brings several key improvements that enhance both the **quality** and **efficiency** of safety assessments:

1. **Tablet-Based Inspections**

Inspectors use tablets to:

- Record observations in real time.
- Capture and embed photos directly into reports.
- Access vessel-specific data instantly.

Benefit: Reduces errors, improves clarity, and speeds up reporting.

2. **Smart Questionnaires**

The **Compiled Vessel Inspection Questionnaire (CVIQ)** is digitally generated based on:

- Vessel type and history.
- Operator performance.
- Risk profile.

Benefit: Ensures inspections are **targeted**, **relevant**, and **risk-based**.

3. Graded and Structured Responses

Digital forms allow for:

- Graded answers (not just yes/no).
- Structured evidence collection.
- Automated consistency checks.

Benefit: Provides deeper insights into safety performance and trends.

4. Data Integration & Analytics

Inspection data is:

- Uploaded to centralized databases.
- Used for benchmarking and predictive safety analysis.

Benefit: Enables **continuous improvement** and **early risk detection**.

5. Human Factors Tracking

Digital tools help assess:

- Crew behavior and competence.
- Performance Influencing Factors (PIFs) like fatigue or morale.

Benefit: Promotes a **no-blame culture** and focuses on systemic safety improvements.

From Conventional To Electric: Advancing Tanker Efficiency And Safety In The IMO 2050 Era

Change is everywhere and expedited need to change is imminent. Technology is in at the helm. Efficiency optimization is the key goal. Efficiency is good for business, it is good for the crew and for the environment we work and live in each day.

Let us embrace and further develop technology to help crew, fleet managers, shipowners and charter parties to secure a safe, enjoyable and economically and environmentally sustainable decision-making